

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_TX_Amarillo-Husband.Amarillo.Intl.AP.723630_TMYx.epw
- Building Name: SmallOffice
- Building Location: Amarillo, TX
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	Wall upgrade: • Exterior walls were upgraded to R-13.0 with Fiberglass Batts insulation. • Wall insulation upgrade completed. Roof upgrade:

		• The roof was upgraded to R-24.4 using Blown Fiberglass insulation. • Roof insulation upgrade completed. Window upgrade: • Upgraded to 3-pane glazing • Window frame: wood window frame • Infiltration reduction: 30% • Glass pane upgrade skipped because SimpleGlazing does not support pane replacement (requested: 3-pane, custom pane count). • Installed window frame: wood window frame. • Applied caulking: acrylic. • Applied glazing film: low-e film. • Applied weatherstripping: silicone adhesive smoke gasket. Door upgrade: • Door type: wooden door • Bottom seal: automatic door bottom • Top/side seals: jamb weatherstrip • Infiltration reduction: 30% • Door retrofit completed.
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Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	195.48	190.41	5.07	2.59%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison

Baseline

\$9,667

Scenario 1

\$9,251

■ Baseline

■ Scenario 1

Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison

Baseline

18,943

Scenario 1

18,466

■ Baseline

■ Scenario 1

Max saving: **\$-416 (-4.30%)**

Max saving: **-478 kg CO₂e (-2.52%)**

Min Retrofit Construction Cost

\$3,795

Scenario 1

Min Retrofit Embodied Carbon

19,345 kg CO₂e

Embodied carbon intensity (ECI): 38 kg CO₂e/m²
Scenario 1

Lowest Retrofit Cost Payback Period

9.12 years

Scenario 1

Lowest Retrofit Carbon Payback Period

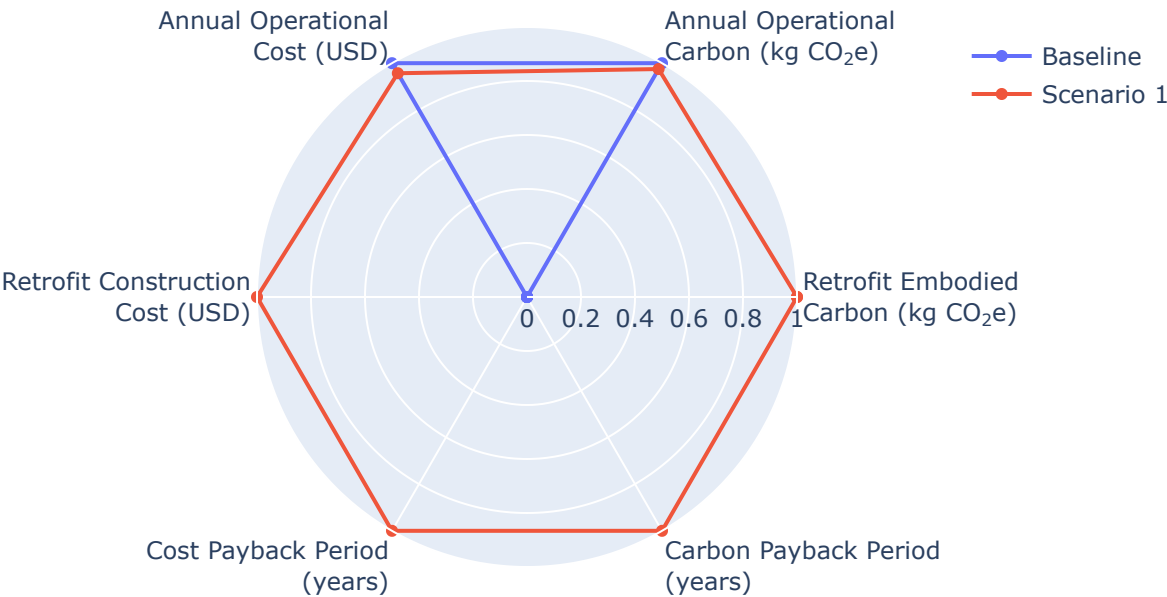
41 years

Scenario 1

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1

9.12 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1

41 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material	Volume (m ³)	Area (m ²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m ³)
Scenario 1	Wall insulation	Fiberglass Batts	N/A	282	N/A	N/A	N/A	N/A
Scenario 1	Roof insulation	Blown Fiberglass	N/A	599	N/A	N/A	N/A	N/A
Scenario 1	Window weatherstripping	silicone adhesive smoke gasket	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 1	Window caulking	acrylic	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 1	Door bottom seal	automatic door bottom	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 1	Door top/side seal	jamb weatherstripping	N/A	N/A	N/A	N/A	N/A	N/A

Result Summary

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	18,943	\$9,667
Scenario 1	19,345	\$3,795	18,466	\$9,251

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